



IDAA

Identity Driven Adversarial Attack

Transfer Attack on Face Verification

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MOTIVATION

Why IDAA?



Identity-aware, not just gradient-driven

An identity-driven adversarial attack, purpose-built to improve transferability across unseen face-verification models — rather than relying purely on output-layer gradients.



New to the baseline

IDAA was not part of the internship's existing attack suite. This work adds it end-to-end: algorithm, registry integration, and evaluation.

CODE REPOSITORY



github.com/Arnav1730/transferattacktInterns-Arnav



METHOD

IDAA — Basic Idea

CORE PRINCIPLE

- Goes beyond final-output gradients by pulling in intermediate feature representations during optimization.
- Perturbations are adapted using intermediate-layer information, producing more transferable adversarial examples.
- Reduces overfitting to the surrogate model and improves black-box attack success.

FEATURE PERTURBATION

$$F = f_l(x) \quad F' = F + N(0, \sigma^2)$$

UPDATE RULE

$$x_{t+1} = \Pi_{\varepsilon} (x_t + \alpha \text{sign}(\nabla_x L))$$

ITERATIVE UPDATE

- 1 Compute image-space gradient
- 2 Apply intermediate feature guidance
- 3 Update adversarial image
- 4 Project into ε -ball
- 5 Clip pixel values

ATTACK OBJECTIVE



Impersonation
Increase cosine similarity



Dodging
Decrease cosine similarity



INTEGRATION

Adapting IDAA to Face Verification



THE CHALLENGE

- IDAA was originally designed for identity-aware feature optimization.
- The baseline framework evaluates CNN-based verification models instead.
- Direct integration into the existing evaluation pipeline was not possible.

MODELS IN THE BASELINE

ArcFace

FaceNet512

GhostFaceNet

VGG-Face



WHAT WE IMPLEMENTED



Identity-aware optimization objective



Intermediate feature-guided perturbation



Iterative gradient-based optimization



Cosine similarity objective



Standard projection constraints (ϵ -ball clipping)



Code Integration

REGISTER IDAA · ATTACK DISPATCHER

```
ALL_ATTACKS = [  
    'PGD', 'MI_FGSM', 'TI_FGSM',  
    'SI_NI_FGSM', 'MI_ADMIX_DI_TI',  
    'BPA_CNN', 'BSR', 'DeCoWA',  
    'SIA_MI_TI', 'OPS', 'ATT_CNN',  
    'LI_BOOST_MI', 'DPA_HMA',  
    'DPA_HMA_ENSEMBLE',  
    'IDAA',  
]  
  
if attack_name == 'IDAA':  
    return idaa(model, src, tgt_emb,  
                attack_type, input_size)
```

New entries highlighted in orange were added to the existing attack registry to wire IDAA into the dispatcher.

idaa_function()

Implementation flow

- 1 Initialize adversarial image
- 2 Generate transformed input (diverse augmentation)
- 3 Compute embedding similarity
- 4 Calculate adaptive iterative gradient
- 5 Update perturbation
- 6 Project into ϵ -ball
- 7 Clip valid pixel range
- 8 Repeat until convergence



Challenges Encountered



CHALLENGE

The attack was not available in the baseline framework.



SOLUTION

Added IDAA support to the attack registry and baseline scripts.



CHALLENGE

No evaluation script existed for IDAA.



SOLUTION

Built an evaluation pipeline generating similarity scores, breach rates, and summary tables.



FINAL STEP

Integrated IDAA into the final comparison table alongside every existing baseline and student attack.



EVALUATION

Final Results



54.17%

Top breach rate — IDAA



+25.0 pts

vs. best baseline (SI-NI-FGSM, 29.17%)



0.287

Highest mean impact score

Attack	Category	Breach Rate (%)	Mean Impact
IDAA	Proposed	54.17	0.287
DPA-HMA	Student	40.21	0.215
BSR	Student	36.46	0.205
LI-BOOST-MI	Student	35.21	0.201
DeCoWA	Student	32.50	0.193
BPA-CNN	Student	30.21	0.180
SI-NI-FGSM	Baseline	29.17	0.176
ATT-CNN	Student	26.67	0.165
MI-FGSM	Baseline	26.67	0.164
MI-ADMIX-DI-TI	Baseline	24.17	0.153
SIA-MI-TI	Student	23.33	0.138
TI-FGSM	Baseline	20.42	0.131
PGD	Baseline	16.67	0.100

CONCLUSION

54.17%

BREACH RATE

0.287

MEAN IMPACT SCORE

IDAA achieved the highest breach rate and mean impact score, outperforming every baseline and student attack evaluated. Among student methods, DPA-HMA, BSR, and LI-BOOST-MI followed with 40.21%, 36.46%, and 35.21%. The strongest baseline, SI-NI-FGSM, reached 29.17%, with MI-FGSM, MI-ADMIX-DI-TI, TI-FGSM, and PGD trailing further behind. IDAA consistently delivers stronger attack success and transferability than existing baseline and student-based approaches.



Thank You

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